VIT-AP University

School of Computer Science and Engineering

B.Tech

MULTIMODAL AGRICULTURAL CHATBOT

An Industrial Internship Report

submitted by

MASUPALLI KUSHWANTH

(23BCE9362)

in partial fulfilment for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering (AI and ML)

March 2026

DECLARATION BY THE CANDIDATE

I hereby declare that the Industrial Internship report entitled " **MULTIMODAL AGRICULTURAL CHATBOT**" submitted by me to VIT-AP University, Amaravati in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering (AI and ML)** is a record of bonafide industrial training undertaken by me under the supervision of the training officer at **Krittya**. I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Name: Masupalli Kushwanth
Reg. No: 23BCE9362

Place: Amaravati
Date: 14 March 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and guided me throughout the course of this internship.

I am deeply thankful to the management and team at Krittya for providing me with the opportunity to work on the Multimodal Agricultural Chatbot project. Their constant guidance, mentorship, and encouragement throughout the internship period were invaluable.

I extend my heartfelt thanks to my academic supervisor and the faculty members of the School of Computer Science and Engineering, VIT-AP University, for their continuous support and constructive feedback during the course of this work.

I am also grateful to VIT-AP University, Amaravati, for providing me with the opportunity to undertake this summer industrial internship as part of my academic curriculum.

This experience has been extremely enriching and has significantly contributed to my technical knowledge and professional development.

Masupalli Kushwanth

Place: Amaravati
Date: 14 March 2026

TABLE OF CONTENTS

Chapter No.	Title	Page No.
—	Abstract	1
1	Introduction	2
1.1	Background	3
2	About the Company	4
3	Problem Statement	5
4	Objectives	6
5	System Overview	7
5.1	System Objectives	7
5.2	System Architecture	7
5.3	Frontend Interface	8
5.4	Modality Detection Module	8
5.5	Speech Processing Pipeline	8
5.6	Video Processing Pipeline	9
5.7	Image Processing Pipeline	9
5.8	Computer Vision Models	9
5.9	Document Processing	10
5.10	NLP Processing Pipeline	10
5.11	Knowledge Retrieval System	11
5.12	Response Generation	11
5.13	Deployment Architecture	12
6	Advantages of the System	12
7	Limitations	13
8	Future Improvements	13
9	Conclusion	14

LIST OF FIGURES

Figure No.	Caption	Page No.
Figure 1	System Architecture of the multimodal Agricultural Chatbot	7

LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
NLP	Natural Language Processing
CNN	Convolutional Neural Network
ML	Machine Learning
API	Application Programming Interface
RAG	Retrieval-Augmented Generation
OCR	Optical Character Recognition
NER	Named Entity Recognition
TTS	Text-to-Speech
STT	Speech-to-Text
MVP	Minimum Viable Product
BERT	Bidirectional Encoder Representations from Transformers
CSE	Computer Science and Engineering

ABSTRACT

Agriculture is a critical sector that supports the livelihood of millions of farmers. However, farmers often face challenges in obtaining timely and reliable information regarding crop diseases, pest control, weather conditions, and modern farming practices. Limited access to agricultural experts and delayed advisory services can negatively impact crop productivity and decision-making.

To overcome these challenges, this project introduces an **AI-based Multimodal Agricultural Chatbot** designed to provide farmers with fast and convenient agricultural assistance. The system allows users to interact through multiple input methods including **text, images, and voice**. Text-based queries are processed using **Natural Language Processing (NLP)** techniques to understand the user's request and generate appropriate responses. For visual analysis, the chatbot uses **Convolutional Neural Networks (CNN)** to detect crop diseases from plant images. In addition, voice-based interaction is supported through the **Sarvam API**, enabling users to communicate with the chatbot using spoken language.

To further improve the effectiveness of the chatbot, a **reinforcement learning-based response selection mechanism** is incorporated. This mechanism helps the system learn from user interactions and improve the relevance of responses over time. The backend of the application is implemented using the **Flask framework**, which handles communication between different system components and manages the integration of machine learning models.

The developed chatbot aims to provide farmers with a **simple, efficient, and accessible platform** for obtaining agricultural guidance quickly, thereby supporting better farming decisions and improving agricultural productivity.

CHAPTER 1: INTRODUCTION

Agriculture plays a significant role in sustaining national economies and supporting the livelihoods of millions of people, particularly in developing nations. Farmers constantly face various challenges such as crop diseases, pest attacks, soil degradation, changing climate conditions, and the need to adopt modern agricultural techniques. Access to accurate and timely information is essential for farmers to manage these challenges effectively and improve crop productivity.

Despite the importance of agricultural advisory services, many farmers find it difficult to obtain expert guidance when needed. Traditional advisory systems often depend on agricultural officers or specialists who may not always be readily available. Geographic limitations, communication delays, and limited resources can further restrict farmers from receiving timely support.

Advancements in **artificial intelligence and digital communication technologies** have created new opportunities to provide automated agricultural assistance. Intelligent conversational systems, commonly known as chatbots, can be used to deliver agricultural knowledge instantly and efficiently. These systems allow users to interact with a virtual assistant that can answer questions and provide recommendations based on user input.

The project presented in this report focuses on the development of an **AI-powered Multimodal Agricultural Chatbot** capable of supporting multiple modes of interaction. The chatbot processes user queries in text form using **Natural Language Processing techniques**, analyzes plant images using **deep learning models based on Convolutional Neural Networks**, and allows users to communicate through **voice input using the Sarvam API**.

Additionally, the system incorporates a **reinforcement learning-based mechanism** that helps improve the quality and relevance of responses through continuous learning from user interactions. The backend of the chatbot is implemented using the **Flask web framework**, which facilitates communication between the system components and ensures efficient processing of user requests.

By combining these technologies, the chatbot aims to provide farmers with an accessible digital tool that can deliver agricultural information quickly and help them make better farming decisions.

1.1 Background

Recent advancements in artificial intelligence and machine learning have enabled the development of intelligent conversational systems capable of assisting users in various domains. Chatbots are widely used in sectors such as healthcare, customer support, and education to provide automated assistance and information retrieval. In agriculture, several research efforts have focused on developing systems that help farmers obtain information related to crop management, pest control, and farming techniques.

Natural Language Processing (NLP) techniques allow chatbots to understand and respond to user queries effectively. In addition, deep learning models such as Convolutional Neural Networks (CNN) have been widely used for image classification tasks, including crop disease detection from leaf images. Voice-based interaction technologies also improve accessibility, especially for users who may find typing difficult.

However, many existing agricultural systems provide limited interaction modes or static responses. Integrating multiple input methods such as text, image, and voice along with adaptive response selection can significantly improve the usefulness and usability of agricultural chatbot systems.

CHAPTER 2: ABOUT THE COMPANY

Krittya is a technology-oriented organization that focuses on developing innovative software solutions using modern technological tools and frameworks. The company actively works on projects involving **artificial intelligence, machine learning, and web-based application development**, aiming to create intelligent systems that solve real-world problems through advanced technology.

The organization emphasizes the practical application of emerging technologies to build scalable and efficient software systems. Through its projects, the company explores various domains where AI-driven solutions can improve productivity, automation, and decision-making.

During the internship period, the company provided valuable guidance and mentorship for the development of the **Multimodal Agricultural Chatbot project**. The internship involved hands-on experience in software development, particularly in backend implementation using the **Flask framework**, integration of machine learning models, and development of intelligent conversational interfaces.

Working on this project provided practical exposure to the complete development lifecycle of an AI-based application. It helped in understanding how machine learning models can be integrated into real-world applications and how backend systems manage communication between different modules.

This internship experience contributed significantly to developing technical skills in **backend development, machine learning integration, and AI-based system design**, while also demonstrating how technology can be used to address real-world challenges such as improving agricultural advisory systems.

CHAPTER 3: PROBLEM STATEMENT

Farmers frequently encounter several challenges while managing their crops, including identifying plant diseases, controlling pests, maintaining soil health, and adapting to changing weather conditions. Obtaining accurate and timely information regarding these issues is essential for ensuring healthy crop growth and improving agricultural productivity.

However, many farmers do not have immediate access to agricultural experts or advisory services. In rural and remote areas, consulting specialists may require significant time and resources. Additionally, traditional advisory systems may not provide instant responses when farmers require urgent assistance.

Due to these limitations, farmers sometimes rely on incomplete or incorrect information when making farming decisions. This can lead to improper use of fertilizers or pesticides, delayed treatment of crop diseases, and ultimately reduced crop yield.

To address these challenges, there is a growing need for an intelligent system that can provide farmers with **quick, reliable, and easily accessible agricultural information**. An AI-powered multimodal agricultural chatbot can serve as such a system by enabling farmers to interact with a digital assistant capable of understanding queries and providing relevant guidance.

By supporting multiple interaction methods such as **text queries, image analysis, and voice communication**, the chatbot can offer farmers an accessible platform for obtaining agricultural advice instantly and conveniently.

CHAPTER 4: OBJECTIVES

The main objectives of this project are:

- To develop an AI-powered chatbot that assists farmers in answering agricultural queries.
- To enable text-based query handling using Natural Language Processing (NLP).
- To implement image-based crop disease detection using Convolutional Neural Networks (CNN).
- To support voice interaction using speech processing through the Sarvam API.
- To improve response quality using a reinforcement learning-based response selection mechanism.
- To integrate all system components using a Flask-based backend.

CHAPTER 5: SYSTEM OVERVIEW

The multimodal agricultural chatbot is designed as a multimodal intelligent system consisting of several interconnected modules that process user inputs in various forms and generate relevant agricultural responses.

5.1 System Objectives

The primary objectives of the multimodal Agricultural Chatbot are:

1. Provide real-time agricultural guidance to farmers.
2. Support multiple forms of input including voice, text, images, videos, and documents.
3. Detect plant diseases, pests, soil types, crops, and weeds using AI models.
4. Enable multilingual communication through automatic language detection and translation.
5. Retrieve information using knowledge-based reasoning and retrieval augmented generation (RAG).
6. Provide responses in both text and speech format.
7. Ensure accessibility through lightweight deployment suitable for MVP testing using FastAPI and ngrok.

5.2 System Architecture

The agricultural chatbot is designed as a **multimodal intelligent system** consisting of several interconnected modules that process user inputs of various types and generate relevant agricultural responses.

Advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and Computer Vision have created opportunities to develop intelligent systems capable of assisting farmers in real time. A Multimodal Agricultural Chatbot provides an interactive platform through which farmers can obtain agricultural advice using multiple input modalities such as text, speech, images, videos, and documents.

The system described in this report implements such a solution by combining the following technologies:

- Speech Processing
- Computer Vision Models
- Natural Language Processing
- Knowledge Retrieval Systems
- Multilingual Translation

The chatbot is capable of understanding agricultural queries, analyzing crop images, identifying pests and weeds, recognizing soil conditions, and retrieving relevant information from a knowledge base. The system architecture integrates several AI modules to provide accurate, multilingual, and accessible agricultural assistance.

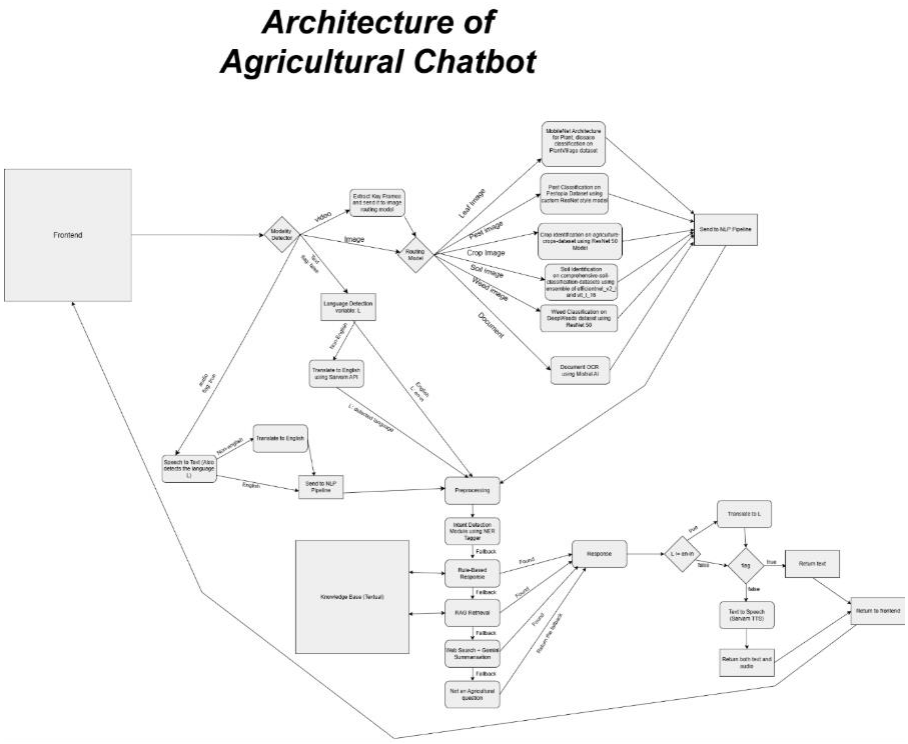


Figure 1: System Architecture of the mulitmodal Agricultural Chatbot

5.3 Frontend Interface

The frontend acts as the user interaction layer of the system. Farmers can interact with the chatbot using text queries, voice messages, image uploads, video uploads, and document uploads. The frontend sends user data to the backend through API endpoints exposed by the FastAPI server.

Since the system is currently in the Minimum Viable Product (MVP) stage, the backend is deployed locally and exposed to the internet using ngrok. This enables remote testing without requiring cloud infrastructure.

5.4 Modality Detection Module

Once the user input reaches the backend, the system first determines the type of input modality. The Modality Detector classifies the incoming data into one of the following categories: text, audio, image, video, or document. This classification determines which processing pipeline will be activated.

5.5 Speech Processing Pipeline

The speech pipeline processes voice queries submitted by the user. The system integrates Sarvam API for speech-related tasks. The speech pipeline operates through the following steps:

8. Speech-to-Text Conversion: The Sarvam API converts spoken audio into text while simultaneously detecting the language of the input. Supported languages include 13 regional languages, making the system accessible to farmers across different linguistic regions.
9. Language Detection: The API detects the language automatically.
10. Translation to English: If the detected language is not English, the system translates the text into English. This standardizes the input so that the NLP pipeline can process it effectively.
11. Forwarding to NLP Pipeline: The translated English text is then sent to the NLP processing module for further analysis.

5.6 Video Processing Pipeline

When users upload videos, the system extracts key frames using scene detection algorithms. Scene detection identifies significant changes in video content and extracts representative frames. These frames are then forwarded to the image processing pipeline. This approach reduces computational overhead while preserving meaningful visual information.

5.7 Image Processing Pipeline

The image pipeline performs computer vision tasks using specialized machine learning models trained for agricultural analysis. After receiving an image input, the system

uses an Image Routing Classifier to determine the type of agricultural content in the image. The classifier identifies whether the image represents plant leaves, pests, crops, soil, or weeds. Once the image type is detected, it is routed to the appropriate machine learning model.

5.8 Computer Vision Models

Multiple specialized models are used to analyze agricultural images.

Plant Disease Detection

Model used: MobileNet Architecture

Dataset: PlantVillage Dataset

MobileNet is chosen due to its lightweight architecture and efficiency for mobile and edge applications. The model identifies plant diseases from leaf images and outputs the predicted disease class.

Pest Classification

Model used: Custom ResNet-based Model

Dataset: Pestopia Dataset

This model identifies different types of crop pests, enabling farmers to recognize infestations early.

Crop Identification

Model used: ResNet50

Dataset: Crop Dataset

The model identifies crop species from images, helping farmers verify crop type or diagnose crop conditions.

Soil Classification

Model used: EfficientNet + Vision Transformer Ensemble

Dataset: Vessel-like Soil Classification Dataset

This ensemble approach combines convolutional feature extraction with transformer-based contextual understanding to improve classification accuracy.

Weed Detection

Model used: ResNet50

Dataset: DeepWeeds Dataset

The model identifies weed species that may affect crop growth.

5.9 Document Processing

For document uploads such as agricultural reports or prescriptions, the system uses Mistral OCR. OCR extracts textual information from documents, which is then forwarded to the NLP pipeline.

5.10 NLP Processing Pipeline

The NLP pipeline processes all textual inputs, whether originating from user text, speech transcription, OCR output, or computer vision predictions. The NLP pipeline consists of the following stages:

1. Preprocessing

Text preprocessing prepares the input for semantic analysis. Typical operations include tokenization, stopword removal, text normalization, and lowercasing.

2. Intent Detection

Intent detection determines what the user wants. The system uses spaCy-based Named Entity Recognition (NER) to identify important agricultural entities such as crop names, pests, diseases, soil types, and fertilizers. This allows the system to classify the query into meaningful agricultural intents.

5.11 Knowledge Retrieval System

The system retrieves information using three levels of response generation.

1. Rule-Based Response System

The rule-based system handles frequently asked questions using predefined responses. Examples include irrigation schedules, fertilizer recommendations, and planting seasons. These responses are stored in a JSON-based knowledge base.

2. Retrieval-Augmented Generation (RAG)

When the rule-based system cannot answer a query, the system uses Retrieval-Augmented Generation (RAG). The process includes converting queries into BERT embeddings, performing semantic similarity search against stored agricultural knowledge documents, retrieving relevant passages, and generating responses based on retrieved information. This improves the chatbot's ability to answer complex questions.

3. Web Search with Gemini Summarization

If the RAG system fails to find relevant information, the chatbot performs a web search. The retrieved web content is summarized using Gemini AI, producing a concise response for the user.

4. Non-Agricultural Query Detection

If the system determines that the query is unrelated to agriculture, it responds appropriately indicating that the question falls outside the domain.

5.12 Response Generation

The response module aggregates outputs from the previous stages and generates the final answer. The response may include disease identification, pest treatment recommendations, crop management advice, and soil improvement suggestions.

Multilingual Response Processing

Since the system supports multiple languages, the final response must be adapted to the user's language. The process involves detecting the original user language and translating the English response back into that language. Translation is performed using Sarvam API.

Text-to-Speech Generation

If the user submitted the query via voice, the system also produces an audio response. Sarvam API is used for Text-to-Speech (TTS) generation. The chatbot returns both a text response and an audio response, which ensures accessibility for users with limited literacy.

5.13 Deployment Architecture

The system backend is implemented using FastAPI, which provides high-performance asynchronous APIs, easy integration with machine learning models, and lightweight deployment. Currently, the system runs on a local server. To allow external access during development and testing, ngrok is used to expose the local API server to the internet. This setup enables rapid testing without requiring full cloud deployment.

CHAPTER 6: ADVANTAGES OF THE SYSTEM

The architecture provides several advantages:

Multimodal Interaction

The system supports voice, text, images, videos, and documents as input modalities, making it highly versatile and accessible for a wide range of users.

Multilingual Support

The chatbot handles 13 regional languages using Sarvam APIs, significantly expanding its reach to farmers across different linguistic regions.

Specialized AI Models

Dedicated models for different agricultural tasks such as plant disease detection, pest classification, soil analysis, and weed detection improve overall accuracy and reliability.

Modular Architecture

Each component of the system can be improved independently without affecting the rest, enabling continuous development and maintenance.

Knowledge Integration

The system effectively combines rule-based systems, RAG retrieval, and web search to handle a diverse range of agricultural queries comprehensively.

CHAPTER 7: LIMITATIONS

Despite its capabilities, the system has certain limitations:

- Local deployment limits scalability for large-scale usage.
- The knowledge base stored in JSON may become inefficient for large datasets.
- Model accuracy depends heavily on training data quality and coverage.
- Video processing may significantly increase computational load on the server.

CHAPTER 8: FUTURE IMPROVEMENTS

Several enhancements can improve the system in future versions:

Cloud Deployment

Deploying the system on cloud infrastructure will improve scalability and allow it to handle a larger number of concurrent users.

Vector Databases

Replacing JSON storage with vector databases such as FAISS or Pinecone will improve retrieval speed and efficiency for large knowledge bases.

Continuous Model Training

Adding new datasets and continuously retraining models will improve classification accuracy across different crop types and regional conditions.

Mobile Application

Developing a dedicated mobile application would significantly improve accessibility for farmers who primarily use smartphones.

Offline Mode

Allowing offline inference on mobile devices would benefit rural users with limited or unreliable internet access.

CHAPTER 9: CONCLUSION

The Multimodal Agricultural Chatbot represents an advanced AI-driven advisory system capable of assisting farmers through natural interaction methods. By integrating speech processing, computer vision, multilingual NLP, and knowledge retrieval mechanisms, the system enables farmers to access agricultural expertise in real time.

The modular architecture allows continuous expansion and improvement, making the chatbot a promising tool for improving agricultural productivity and reducing crop losses. The current MVP implementation demonstrates the feasibility of such a system and lays the foundation for future large-scale agricultural AI platforms.

This internship at Krittya provided an invaluable opportunity to apply theoretical knowledge in a practical setting, developing skills in backend development, machine learning integration, and AI-based system design. The experience gained through this project will serve as a strong foundation for future professional development in the field of artificial intelligence and software engineering.

APPENDIX 1: INTERNSHIP DETAILS

Student Name	Masupalli Kushwanth
Registration Number	23BCE9362
Role	SDE Intern
Company	Krittya
Internship Duration	1 January 2026 – 1 March 2026
University	VIT-AP University
Department	CSE
Branch	AI and ML
Date of Submission	14 March 2026